In [ ]:
```
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#Roll No : A33
#Batch : A - 2
#Assignment No - 08 [EDA Experiment -> Feature Selection]
```

In [5]:
```
import numpy as np
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier
from sklearn.feature_selection import SequentialFeatureSelector
```

In [7]:
```
# Load Iris dataset
iris = load_iris()
X = pd.DataFrame(iris.data, columns=iris.feature_names)
y = pd.Series(iris.target, name='target')
df = pd.concat([X, y], axis=1)
df
```

Out[7]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	2
146	6.3	2.5	5.0	1.9	2
147	6.5	3.0	5.2	2.0	2
148	6.2	3.4	5.4	2.3	2
149	5.9	3.0	5.1	1.8	2

150 rows × 5 columns

In [8]:
```
# Filter Methods (Correlation)
# Q.1 Using the Iris dataset,
# Apply Correlation and display the selected features and their ranking.
correlations = df.corr()['target'].drop('target')

corr_summary = pd.DataFrame({
    'Feature': correlations.index,
    'Correlation': correlations.values,
    'Absolute Correlation': correlations.abs().values
```

```
}).sort_values(by='Absolute Correlation', ascending=False).reset_index(dr

corr_summary['Rank'] = corr_summary.index + 1
print("--- Feature Ranking via Correlation ---")
print(corr_summary[['Rank', 'Feature', 'Correlation', 'Absolute Correlati
```

```
--- Feature Ranking via Correlation ---
   Rank            Feature  Correlation  Absolute Correlation
0     1   petal width (cm)     0.956547              0.956547
1     2  petal length (cm)     0.949035              0.949035
2     3  sepal length (cm)     0.782561              0.782561
3     4   sepal width (cm)    -0.426658              0.426658
```

In [9]:
```
# Wrapper Methods (Sequential Feature Selection)
# Q2.  Using the Iris dataset,
# • Apply Sequential Backward Selection with DecisionTree Classifier to s
# the most important features.
# • Display the selected features.
dt_classifier = DecisionTreeClassifier(random_state=42)

sbs = SequentialFeatureSelector(
    estimator=dt_classifier,
    n_features_to_select=2,
    direction='backward',
    cv=5,
    scoring='accuracy'
)
sbs.fit(X, y)

sbs_selected = X.columns[sbs.get_support()].tolist()
print("Selected Features (Sequential Backward Selection):", sbs_selected)
```

```
Selected Features (Sequential Backward Selection): ['petal length (cm)',
'petal width (cm)']
```

In [10…
```
# Q3. Using the Iris dataset,
# • Apply Sequential Forward Selection (SFS) with DecisionTree Classifier
# select the most important features.
# • Display the selected features.
# Sequential Forward Selection
sfs = SequentialFeatureSelector(
    estimator=dt_classifier,
    n_features_to_select=2,
    direction='forward',
    cv=5,
    scoring='accuracy'
)
sfs.fit(X, y)

sfs_selected = X.columns[sfs.get_support()].tolist()
print("Selected Features (Sequential Forward Selection):", sfs_selected)
```

```
Selected Features (Sequential Forward Selection): ['petal length (cm)',
'petal width (cm)']
```

In [11…

```python
# Embedded Methods (e.g., DecisionTree)
# Q4.
# • Train a DecisionTree Classifier on the Iris dataset and obtain the fe
# importances.
# • List the features in order of importance and explain why some feature
# might be more important than others.

# Train Decision Tree
tree = DecisionTreeClassifier(random_state=42)
tree.fit(X, y)

# Extract and rank feature importances
importance_df = pd.DataFrame({
    'Feature': X.columns,
    'Importance': tree.feature_importances_
}).sort_values(by='Importance', ascending=False).reset_index(drop=True)

importance_df['Rank'] = importance_df.index + 1
print("--- Decision Tree Feature Importances ---")
print(importance_df[['Rank', 'Feature', 'Importance']])
```

```
--- Decision Tree Feature Importances ---
   Rank            Feature  Importance
0     1  petal length (cm)    0.564056
1     2   petal width (cm)    0.422611
2     3  sepal length (cm)    0.013333
3     4   sepal width (cm)    0.000000
```